Slides 7,8

$$NPV = \sum_{t=1}^{n} \frac{C_t}{(1+r)^t}$$ → If NPV → +ve → accept the project

" " → -ve → reject

↳ Disadvantage → does not consider life of the project and scale of initial investment

Benefit Cost Ratio (BCR)

$$BCR = \frac{PVB}{I}$$ → considers initial investment

NBCR (Net Benefit Cost Ratio) $= \frac{PVB - I}{I} = BCR - 1$

Internal Rate of Return (IRR) → can give multiple rates (disadvantage) ①

Discount rate which makes NPV = 0 ; The discount rate that makes NPV = 0 is also the rate of return

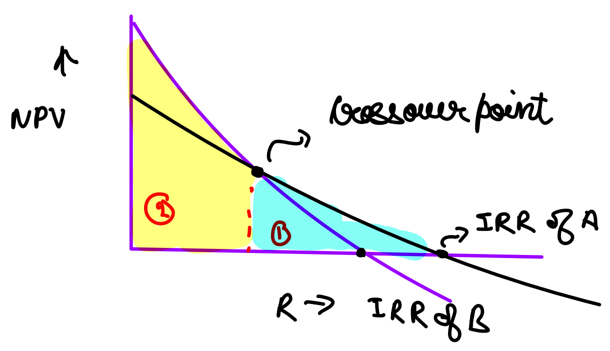
$$-C_0 + \frac{C_1}{1+IRR} + \frac{C_2}{(1+IRR)^2} + \frac{C_3}{(1+IRR)^3} \cdots \frac{C_n}{(1+IRR)^n} = 0$$

Acceptance Rule
The firm should accept an investment project if the opportunity cost of capital is less than the internal rate of return

→ i.e. accept if IRR of your project > cost of capital

i.e. your project gives better returns than opportunity cost of capital

② IRR may give misleading results for two different mutually exclusive events / projects



In zone ①, after cross-over point → (yellow)
$NPV_A < NPV_B$

" " ② before " " " → (blue)
$NPV_B < NPV_A$

But IRR says → invest in A, whereas $NPV_B > NPV_A$ in zone 1

③ Unreliable for different patterns of cash flows

Project	C_0	C_1	C_2	C_3	C_4	IRR	NPV at 10%
X	−110,000	+31,000	+40,000	+50,000	+70,000	22%	36,613
Y	− 110,000	+71,000	+40,000	+40,000	+20,000	25%	31,314

cost of capital of 10 per cent.

IRR of Y > IRR of X, but NPV of X > NPV of Y → IRR misleading

Profitability Index $= \dfrac{NPV}{Investment}$

Pros	Cons
■ Closely related to NPV	■ May lead to multiple rates of return
■ Easy to understand and interpret	■ May result into incorrect decisions in comparing mutually exclusive projects

Modified Internal Rate of Return (MIRR)

$$PVC = \frac{TV}{(1+MIRR)^n}$$

→ Take terminal value of cash inflows
→ Take present value of cash outflow / expenses

Example of MIRR

Year	0	1	2	3	4	5	6
Cash flow	−120	− 80	20	60	80	100	120

cost of capital is 15 percent.

The present value of costs is:

$$120 + \frac{80}{(1.15)} = 189.6$$

The terminal value of cash inflows is:

$$20(1.15)^4 + 60(1.15)^3 + 80(1.15)^2 + 100(1.15) + 120$$
$$= 34.98 + 91.26 + 105.76 + 115 + 120 = 467$$

The MIRR is obtained as follows:

$$189.6 = \frac{467}{(1+MIRR)^6}$$
$$(1 + MIRR)^6 = 2.463$$
$$1 + MIRR = 2.463^{1/6} = 1.162$$
$$MIRR = 1.162 - 1 = 0.162 \text{ or } 16.2 \text{ percent}$$

Book / Accounting Rate of Return

$$\text{Book rate of return} = \frac{\text{book income}}{\text{book assets}}$$

$$\text{Avg accounting return} = \frac{PAT}{\text{Book value of investment}} = \frac{\text{Avg. Net Income}}{\text{Avg. book value}}$$

Disadvantages
→ Doesn't take in account TVM
→ Based on accounting profit, not cash flow
→ Accounting Rate of return doesn't provide on what target rate of return should be

Payback period
↳ length of time required to recover initial investment / cash outlay

ADVANTAGE AND DISADVANTAGE

- It is simple, both in concept and application.
- It favours projects which generate substantial cash inflows in earlier years and discriminates against projects which bring substantial cash inflows in later years but not in earlier years.
- Since it emphasises earlier cash inflows, it may be a sensible criterion when the firm is pressed with problems of liquidity.

- It fails to consider the time value of money. Cash inflows, in the payback calculation, are simply added without suitable discounting.
- It ignores cash flows beyond the payback period. This leads to discrimination against projects which generate substantial cash inflows in later years.

Discounted payback period → Takes in consideration of TVM, somewhat similar to "duration"
↳ time duration until sum of discounted cash flow is equal to initial investment

Slide 8

A firm's choice of how much debt it should have relative to equity is known as capital structure decision

Sources of funds for corporations
- Debt — loan, bond, debentures
- Equity
- Preference share

Cost of equity
↳ represents the compensation that the market demands in exchange for owning the asset and bearing risk of ownership

How to calculate?

SML / CAPM — Dividend growth model

Dividend growth model approach

$$P_0 = \frac{D_0 \times (1+g)}{r-g} = \frac{D_1}{r-g} \Rightarrow R_E = \frac{D_1}{P_0} + g$$

R_E → required return

g = retention ratio (b) × return on equity (ROE)

$g = b \times ROE$

$b = 1 -$ Dividend payout ratio

$= 1 - DPR$

ADVANTAGES AND DISADVANTAGES OF DIVIDEND GROWTH MODEL

- Advantage – easy to understand and use
- Disadvantages
  - Only applicable to companies currently paying dividends
  - Not applicable if dividends aren't growing at a reasonably constant rate
  - Extremely sensitive to the estimated growth rate – an increase in g of 1% increases the cost of equity by 1%
  - Does not explicitly consider risk

SML Approach

$$E(R_e) = R_f + \beta_e \times [E(R_m) - R_f]$$

$$R_e = R_f + \beta[R_m - R_f]$$

R_f → risk free rate

$E(R_m) - R_f$ → market risk premium

β → systematic risk

ADVANTAGES AND DISADVANTAGES OF SML

- Advantages
  - Explicitly adjusts for systematic risk
  - Applicable to all companies, as long as we can estimate beta
- Disadvantages
  - Have to estimate the *expected* market risk premium, which does vary over time
  - Have to estimate beta, which also varies over time
  - We are using the past to predict the future, which is not always reliable.

Cost of Debt - best estimated by computing yield to maturity on existing debt

↳ cost of debt is NOT coupon rate

Cost of preference stock → rate of return required by investors on preferred stocks issued by the company

$$R_P = \frac{D_P}{P_{net}}$$

WACC (Weighted average cost of capital)

D → market value of debt

E → " " " equity

P → " " " of prefered stocks

V → " " the firm = D+E+P

$$WACC = W_E R_E + W_D R_D (1-T_c)$$

$$WACC = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D \times (1-T_c) + \left(\frac{P}{V}\right) \times R_P \qquad [V = D+E+P]$$

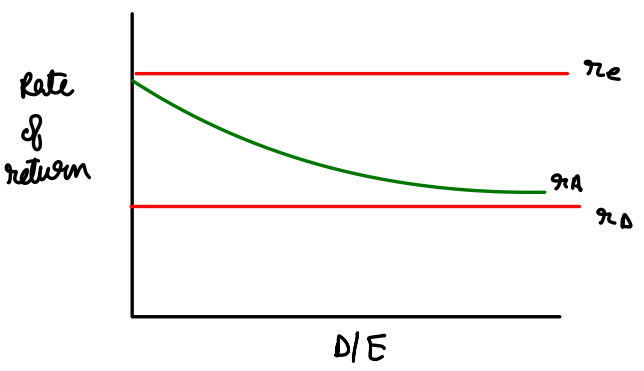
CAPITAL STRUCTURE

1. Net Income approach
2. Net operating Income approach
3. Traditional approach
4. M&M theory of capital structure

↳ Proposition I – Firm Value

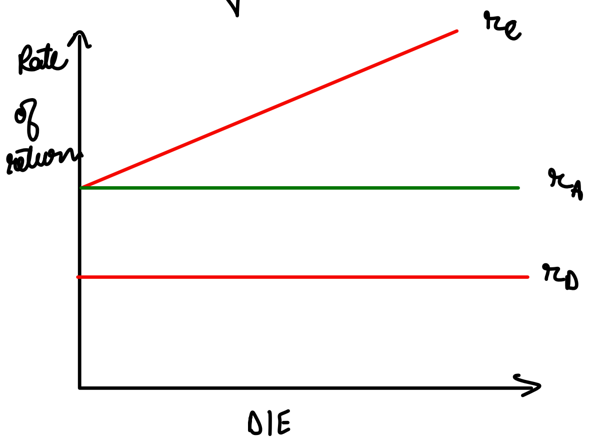
Proposition II – Cost of capital

Net Income Approach



- Increase in debt won't affect confidence level of investors
- Only two sources of finance – debt, equity
- No flotation cost, no transaction cost, no corporate tax
- Capital market is perfect
  - → No information assymetry
  - → Investors are rational

Net operating Income Approach



- Assume $r_A \rightarrow$ overall capitalisation cost & $r_D \rightarrow$ cost of debt to remain same.

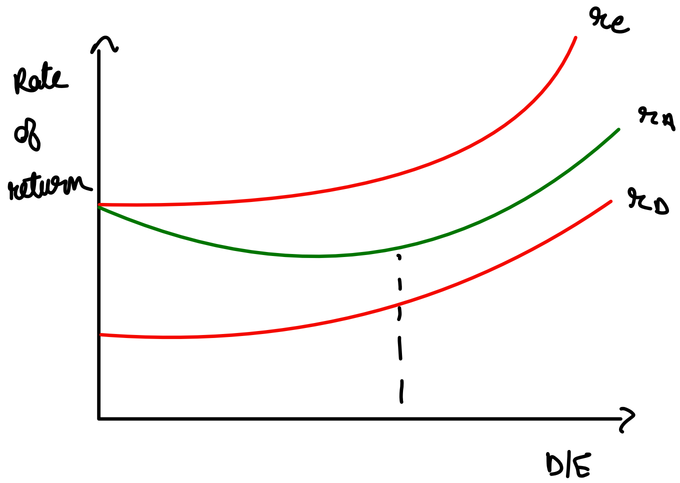
$$r_A = \left(\frac{D}{D+E}\right) \times r_D + \left(\frac{E}{D+E}\right) \times r_e$$

$$r_e \times \left(\frac{E}{D+E}\right) = r_D \times \left(\frac{D}{D+E}\right) - r_A$$

$$= r_D \times \frac{D}{E} - r_A \times \frac{(D+E)}{E}$$

$$r_e = \frac{D}{E}(r_D - r_A) - r_A$$

Traditional Approach

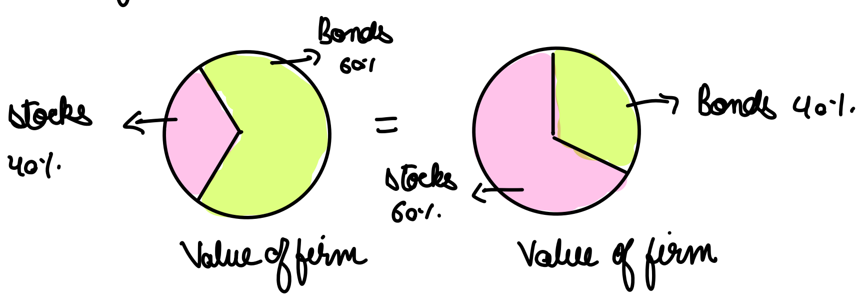


- $r_D \rightarrow$ remains more or less constant but rises thereafter at an increasing rate
- r_e – " " " "
- r_A – (i) decreases upto certain point
  (ii) remains more or less same
  (iii) rises beyond a certain point

MM propositions

M&M Proposition I: The pie model

$\rightarrow$ Value of a firm is independent of firm's capital structure $\rightarrow$ assuming operating earnings remain same



M&M proposition II

$\rightarrow$ Cost of equity increases linearly as company increases its proportion of debt financing

$$R_E = \underbrace{R_A}_{\text{captures business risk}} + (R_A - R_D) \times \frac{D}{E} \Big\} \text{ captures financial risk}$$

captures business risk
$\hookrightarrow$ depends on firm's operating activities

Value of Tax Shield

$$V_L = V_U + T_C \times D$$

M&M II with corporate taxes :

$$R_E = R_A + (R_A - R_D) \times \frac{D}{E} \times (1 - T_C)$$